

Unit 5

Data Management and Business Intelligence

(4 Hours)

5.1 Importance of data in decision-making

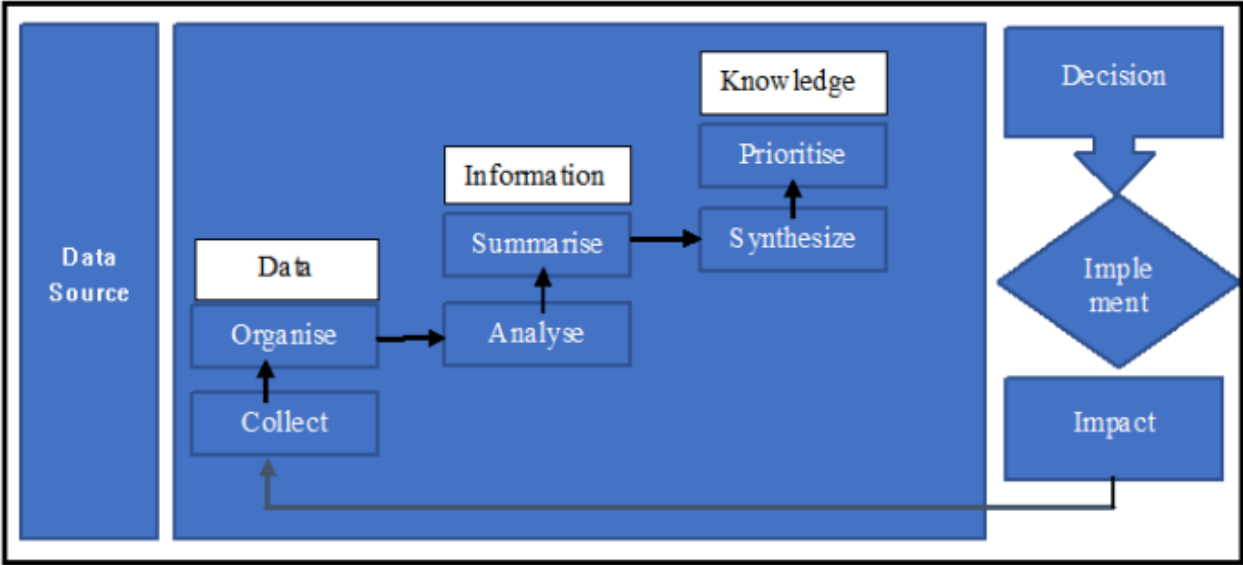
5.2 Data warehousing and data mining

5.3 Business intelligence tools and techniques

5.4 Big data analytics and its business applications

Objectives:

- *Analyze how databases can be used for competitive advantage.*
- *Understand the relationship between business strategy IS strategy and business intelligence*



In Management Information System, raw data is collected from different business sources. Then it is organized, analyzed, summarized, synthesized, and prioritized. Finally, managers use it to make decisions. After implementation, the impact is again observed and sent back as feedback for future decisions.

Data

Data means raw facts, figures, and observations collected from business activities.

Data-Driven Decision-Making (**DDDM**) is the process of using verified facts, metrics, and data analysis to guide strategic decisions rather than relying purely on intuition or guesswork.

It enables organizations to minimize risk, reduce personal bias, and identify new operational opportunities

1. Meaning of Data in MIS

Data means raw facts collected from business operations.

Example:

Data Source	Raw Data
Sales system	Daily sales amount
HR system	Staff attendance

Data Source	Raw Data
Inventory system	Stock quantity
Finance system	Expenses and income
Customer system	Complaints and feedback
Website/App	Clicks, orders, reviews

Raw data itself may not directly support decision-making. It must be processed.

2. Explanation of the Diagram

Step 1: Data Source

Data source is the place where data is generated.

Examples:

Data Source	Example
POS billing system	Sales bills
College MIS	Attendance and marks
Hospital MIS	Patient records
Banking system	Transaction records
E-commerce website	Orders and clicks

The first MIS activity is to collect data from different sources.

Example in a retail store:

Date	Product	Quantity Sold	Unit Price
1 Jan	Laptop	3	70,000
1 Jan	Mouse	20	700
1 Jan	Keyboard	10	1,200

This is raw sales data.

Step 3: Organize Data

Collected data must be arranged properly.

Example:

Instead of random records, MIS organizes data by:

Date	Branch	Product	Category	Quantity	Price
1 Jan	Butwal	Laptop	Computer	3	70,000
1 Jan	Butwal	Mouse	Accessory	20	700
1 Jan	Bhairahawa	Keyboard	Accessory	10	1,200

Organized data is easier to search, filter, and analyze.

Analysis means finding meaning, pattern, or relationship from data.

Add calculated column:

Product	Quantity	Unit Price	Sales
Laptop	3	70,000	210,000
Mouse	20	700	14,000
Keyboard	10	1,200	12,000

Analysis:

- Laptop has the highest revenue.
 - Mouse has the highest quantity sold.
 - Accessories sell frequently but give lower revenue.
-

Step 5: Summarize Information

After analysis, MIS summarizes data into useful information.

Category	Total Sales
Computer	210,000
Accessory	26,000

Now the manager can quickly understand performance.

This is **information**, not just raw data.

Step 6: Synthesize Knowledge

Synthesis means combining different information to create deeper understanding.

Sanjeev Thapa, Er. DevOps, SRE, CKA, RHCSA, RHCE, RHCSA-Openstack, MTCNA, MTCTCE, UBSRS, HEv6, Research Evangelist

Information	Knowledge
Laptop gives high revenue	Premium products are important
Mouse sells in high quantity	Accessories have high demand
Accessory sales are frequent	Bundle offer can increase sales

Knowledge:

Customers buying laptops may also need mouse, keyboard, bag, and antivirus.

Step 7: Prioritize Knowledge

Managers must decide which issue is more important.

Example:

Issue	Importance	Priority
Laptop stock is low	High revenue product	High
Mouse stock is low	High quantity product	Medium
Keyboard sales are low	Less urgent	Low

MIS helps managers focus on important business issues first.

Step 8: Decision

Based on knowledge, management makes decisions.

Example decisions:

- Increase laptop stock.

- Create laptop + mouse + keyboard bundle.
- Give discount on slow-moving keyboard.
- Target corporate customers for laptop sales.

Step 9: Implementation

Decision must be applied in real business.

Example:

Decision	Implementation
Increase laptop stock	Order 20 more laptops
Bundle offer	Laptop + mouse + bag package
Promote accessories	Display accessories near billing counter
Target corporate buyers	Send quotation to offices

Step 10: Impact

After implementation, MIS measures impact.

Indicator	Before Decision	After Decision
Laptop sales	3 units/day	5 units/day
Accessory sales	Rs. 26,000/day	Rs. 45,000/day
Customer satisfaction	75%	88%
Profit	Rs. 40,000/day	Rs. 65,000/day

If the impact is positive, the decision was successful.

If the impact is negative, management must revise the decision.

Step 11: Feedback

The impact becomes new data for future decisions.

Example:

After bundle offer, accessory sales increased. This result becomes new feedback.

So MIS decision-making is a continuous cycle:

Data → Information → Knowledge → Decision → Action → Impact → Feedback → Better Decision

Example 1: College MIS Decision-Making

A college uses MIS to manage students.

Raw Data

Student	Attendance %	Assignment Score	Internal Marks	Fee Status
Ram	92	18	35	Paid
Sita	65	12	24	Due
Hari	78	15	30	Paid
Gita	55	8	18	Due
Nabin	88	17	34	Paid

Analysis

Student	Risk Level	Reason
Ram	Low	Good attendance and marks
Sita	Medium	Low attendance, fee due
Hari	Low	Acceptable performance
Gita	High	Low attendance, low marks, fee due
Nabin	Low	Good performance

MIS Information

- 2 students have attendance problems.
- 1 student is at high academic risk.
- 2 students have fee dues.
- Low attendance is related to low internal marks.

Management Decision

Problem	Decision
Low attendance	Call student and parents
Low internal marks	Provide extra tutorial class
Fee due	Send reminder notice
High-risk student	Academic counseling

Impact

Indicator	Before	After
Average attendance	75.6%	84%
Assignment submission	70%	90%
Internal pass rate	80%	95%

This shows how MIS supports academic decision-making.

4. Example 2: Hospital MIS Decision-Making

A hospital collects patient flow data.

Raw Data

Department	Daily Patients	Average Waiting Time	Doctors Available
OPD	180	90 min	3
Emergency	60	20 min	4
Lab	120	70 min	2
Pharmacy	200	50 min	2

Analysis

Department	Problem
OPD	High patient load and long waiting time
Lab	Long waiting time

Department	Problem
Pharmacy	Many patients but few counters

MIS Information

- OPD has the highest waiting time.
- Pharmacy has highest service volume.
- Lab needs faster reporting.
- Staff allocation is not balanced.

Decision

Issue	Decision
OPD waiting time high	Add one doctor during peak hours
Lab delay	Add one lab technician
Pharmacy queue long	Open one extra billing counter
Patient dissatisfaction	Display token system

Impact

Indicator	Before	After
OPD waiting time	90 min	45 min
Lab waiting time	70 min	35 min
Pharmacy waiting time	50 min	25 min

Indicator	Before	After
Patient satisfaction	68%	86%

MIS helps hospital management improve service quality.

5. Example 3: E-Commerce MIS Decision-Making

An online store collects customer behavior data.

Raw Data

Customer	Page Views	Added to Cart	Purchased	Amount
C1	12	Yes	Yes	4,500
C2	9	Yes	No	0
C3	20	Yes	Yes	8,000
C4	5	No	No	0
C5	15	Yes	No	0

Analysis

Metric	Value
Total visitors	5
Added to cart	4
Purchased	2

Metric	Value
Cart abandonment	2
Conversion rate	40%

MIS Information

- Many customers add products to cart but do not buy.
- Cart abandonment is high.
- Customers with high page views are more interested.
- Checkout process may be difficult.

Decision

Problem	Decision
Cart abandonment	Send reminder message
Low conversion	Offer 5% discount
Checkout issue	Simplify payment process
Interested customers	Retarget through ads

Impact

Indicator	Before	After
Conversion rate	40%	55%
Cart abandonment	50%	30%

Indicator	Before	After
Sales	Rs. 12,500	Rs. 22,000

6. Example 4: Inventory MIS Decision-Making

A shop maintains inventory data.

Raw Data

Product	Current Stock	Monthly Demand	Reorder Level
Rice	100 kg	300 kg	80 kg
Oil	40 liter	200 liter	60 liter
Sugar	90 kg	150 kg	50 kg
Tea	25 kg	100 kg	40 kg

Analysis

Product	Status
Rice	Safe
Oil	Reorder needed
Sugar	Safe
Tea	Reorder needed

MIS Information

- Oil stock is below reorder level.
- Tea stock is below reorder level.
- Rice and sugar stock are enough.

Decision

Product	Action
Oil	Order 200 liters
Tea	Order 100 kg
Rice	No immediate order
Sugar	No immediate order

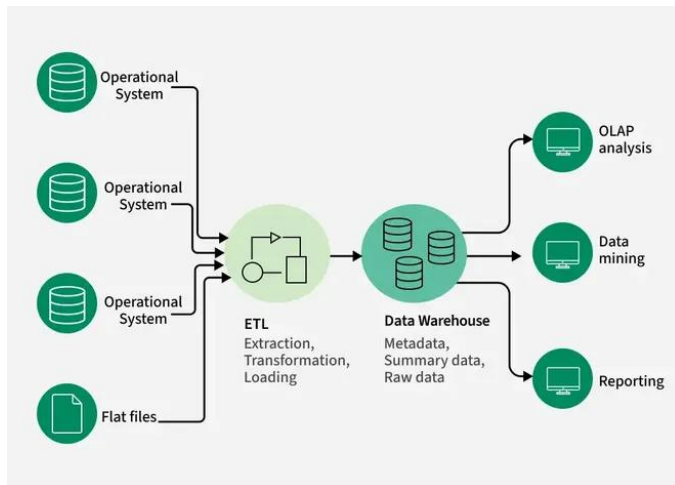
Impact

- Stock-out is avoided.
- Customer demand is fulfilled.
- Emergency purchase cost is reduced.
- Inventory planning improves.

A. Data Warehousing

1. Meaning of Data Warehouse

A **data warehouse** is a central storage system where data from different departments or systems is collected, cleaned, integrated, and stored for analysis and reporting.



Simple Definition

A data warehouse is a large, centralized database that stores historical and integrated data from different sources to support managerial decision-making.

Example:

A college has different systems:

System	Data
Admission system	Student profile, entrance score
Attendance system	Class attendance
Exam system	Marks, grades

System	Data
Finance system	Fee paid, fee due
LMS system	Login, assignment, quiz activity

A **data warehouse** combines all these data into one place so management *can analyze student performance, fee collection, dropout risk, and academic quality.*

2. Why Data Warehouse Is Needed

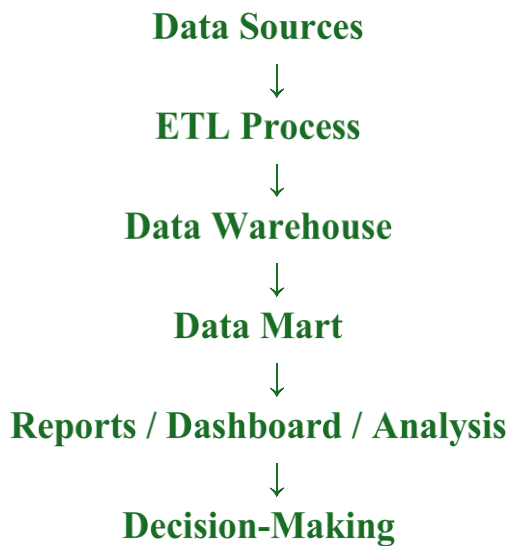
In many organizations, data is scattered in different systems.

Example:

Department	Separate Data Problem
Sales	Sales data in Excel
Finance	Expense data in accounting software
HR	Staff data in HR system
Inventory	Stock data in inventory software
Customer service	Complaints in CRM

If the manager wants a complete business picture, separate systems are not enough.

A data warehouse solves this problem by creating a *single integrated data platform.*



4. ETL in Data Warehouse

ETL means:

ETL Step	Meaning	Example
Extract	Collect data from different sources	Sales, finance, HR, inventory
Transform	Clean and convert data into standard format	Remove duplicate records
Load	Store cleaned data into warehouse	Load into BI database

The identifies ETL and data integration as important parts of BI architecture, because data must be extracted from operational systems, transformed, and loaded into the warehouse for analysis.

Raw Data Before Cleaning

Student Name	Phone	Program	Attendance
Ram Thapa	9847000000	BCA	82
R. Thapa	9847000000	BCA	82
Sita KC	9857000000	BCSIT	76
Hari Sharma	9867000000	bca	68

Problems:

- Duplicate record: Ram Thapa and R. Thapa
- Program format inconsistency: BCA and bca
- Attendance below required level for Hari

After Transformation

Student ID	Student Name	Phone	Program	Attendance %
S001	Ram Thapa	9847000000	BCA	82
S002	Sita KC	9857000000	BCSIT	76
S003	Hari Sharma	9867000000	BCA	68

Now data is clean and ready for analysis.

1. Subject-Oriented

Data warehouse is organized around major subjects.

Examples:

Subject	Data Stored
Sales	product, customer, revenue
Student	attendance, marks, fee
Finance	income, expense, profit
HR	staff, salary, performance

2. Integrated

Data from different sources is combined in one standard format.

Example:

Source	Date Format
Sales system	05/21/2026
Finance system	2026-05-21
Excel sheet	21-May-2026

In the warehouse, all dates are stored in one format:

Standard Date

2026-05-21

3. Time-Variant

A data warehouse stores historical data.

Example:

Year	Sales
2022	1,500,000
2023	2,000,000
2024	2,800,000
2025	3,500,000

This helps trend analysis and forecasting.

4. Non-Volatile

Once data is stored in a warehouse, it is usually not changed frequently. It is mainly used for analysis.

Example:

Last year's sales record is stored permanently for comparison.

6. Data Warehouse vs Operational Database

Basis	Operational Database	Data Warehouse
Purpose	Daily transaction processing	Analysis and decision-making
Users	Staff, operators	Managers, analysts
Data	Current data	Historical data
Operation	Insert, update, delete	Query, report, analyze
Example	Fee payment entry	Fee collection trend report
System Type	OLTP	OLAP

7. OLTP and OLAP

OLTP: Online Transaction Processing

OLTP handles daily transactions.

Examples:

- *Billing*
- *Admission entry*
- *Fee payment*
- *Product sale*
- *Salary entry*

OLAP analyzes data from different dimensions.

Examples:

- *Sales by month*
- *Fee collection by program*
- *Attendance by semester*
- *Profit by branch*

OLTP	OLAP
Transaction focused	Analysis focused
Current data	Historical data
Used by clerks and staff	Used by managers
Example: record a sale	Example: analyze yearly sales

8. Data Mart

A **data mart** is a smaller part of a data warehouse, focused on a specific department or subject.

The data warehouse may consist of one or more **data marts**, and a data mart focuses on a single subject or line of business.

Data Mart	Used By
Sales data mart	Sales manager
Finance data mart	Finance manager

Data Mart	Used By
HR data mart	HR department
Student performance data mart	Academic coordinator
Inventory data mart	Store manager

Example

A college data warehouse may contain:

College Data Warehouse

- Admission Data Mart
- Attendance Data Mart
- Exam Data Mart
- Finance Data Mart
- LMS Data Mart

Data Warehouse Table

Student	Program	Attendance %	Internal Marks	Fee Status	LMS Login/Week	Final Result
Ram	BCA	92	38	Paid	5	Pass
Sita	BCA	64	22	Due	1	Fail
Hari	BCSIT	80	34	Paid	4	Pass
Gita	BCSIT	55	18	Due	0	Fail
Nabin	BCA	88	36	Paid	6	Pass

9. Example: College Data Warehouse

Data Sources

Source	Data
Admission	GPA, entrance score, program
Attendance	attendance percentage
Exam	assignment, internal marks, final marks
Finance	fee paid, fee due
LMS	login count, quiz activity

Analysis from Warehouse

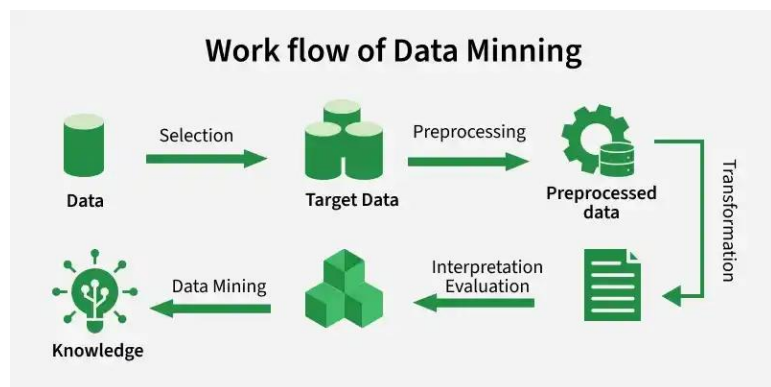
Question	Answer
Which students are at risk?	Sita and Gita
Which factor affects result?	Low attendance and low LMS activity
Which program needs support?	Compare BCA and BCSIT result
Which students need counseling?	Fee due + low marks students

- Conduct extra classes for weak students.
- Contact students with low attendance.
- Send fee reminder to due students.
- Use LMS activity as early warning indicator.

B. Data Mining

1. Meaning of Data Mining

Data mining is the process of discovering hidden patterns, relationships, trends, and useful knowledge from large datasets.



Simple Definition

Data mining means digging useful knowledge from data.

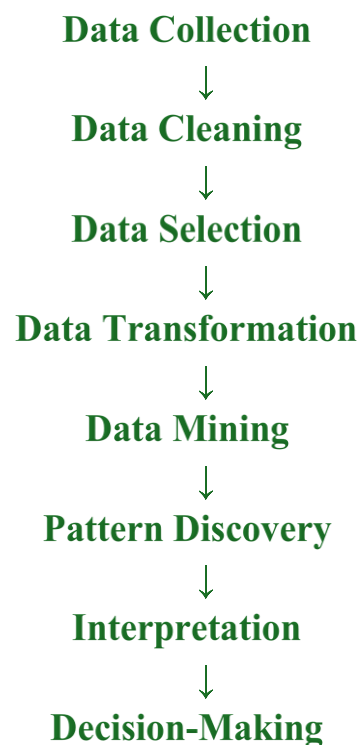
Just like mining gold from earth, data mining extracts valuable insights from large data.

2. Data Mining in MIS

In MIS, data mining helps managers answer questions such as:

Business Question	Data Mining Output
Which customers may leave?	Churn prediction
Which students may fail?	Academic risk prediction
Which product sells together?	Association pattern
Which transaction is suspicious?	Fraud detection
Which branch performs best?	Performance grouping
What will be next month's sales?	Forecasting

3. Data Mining Process



4.1 Classification

Classification assigns data into predefined categories.

Example: Student Risk Classification

Student	Attendance %	Internal Marks	LMS Login/Week	Category
Ram	92	38	5	Low Risk
Sita	64	22	1	High Risk
Hari	80	34	4	Low Risk
Gita	55	18	0	High Risk
Nabin	72	26	2	Medium Risk

Decision:

- *High-risk students need counseling.*
- *Medium-risk students need monitoring.*
- *Low-risk students continue normally.*

Use cases:

- *Loan approval*
- *Spam detection*
- *Student pass/fail prediction*
- *Customer loyal/not loyal*

Clustering groups similar records together without predefined categories.

Example: Customer Segmentation

Customer	Monthly Spending	Visit Frequency	Cluster
C1	50,000	8	Premium Customer
C2	5,000	2	Low-Value Customer
C3	30,000	6	Regular Customer
C4	4,000	1	Low-Value Customer
C5	70,000	10	Premium Customer

Decision:

Cluster	Business Action
Premium customer	Give loyalty card
Regular customer	Offer bundle package
Low-value customer	Send discount campaign

4.3 Association Rule Mining

Association rule mining finds relationships between items.

Example: Retail Basket Analysis

Customer Items Bought

C1 Rice, Oil, Lentil

C2 Rice, Oil

C3 Bread, Butter

C4 Rice, Oil, Sugar

C5 Bread, Butter, Jam

Pattern:

- Customers who buy **rice** often buy **oil**.
- Customers who buy **bread** often buy **butter**.

Decision:

- Place rice and oil nearby.
- Create rice + oil combo offer.
- Place bread and butter together.

4.4 Regression

Regression predicts numerical values.

Example: Advertisement and Sales

Advertisement Cost	Sales
10,000	80,000
20,000	150,000
30,000	230,000
40,000	310,000

Possible prediction:

If advertisement cost is Rs. 50,000, expected sales may be around Rs. 390,000.

Use cases:

- *Sales forecasting*
- *Demand prediction*
- *Profit prediction*
- *Price estimation*

4.5 Anomaly Detection

Anomaly detection finds unusual or suspicious records.

Example: Banking Transactions

Transaction	Amount	Location	Status
T1	2,000	Butwal	Normal
T2	3,500	Butwal	Normal
T3	1,800	Bhairahawa	Normal

Transaction	Amount	Location	Status
T4	250,000	Unknown	Suspicious

Decision:

- *Block transaction temporarily.*
- *Send OTP verification.*
- *Alert fraud monitoring team.*

4.6 Forecasting

Forecasting predicts future values using past data.

Example: Monthly Admission Inquiry

Month	Inquiries
January	80
February	100
March	130
April	160
May	200

Possible decision:

- Increase admission counselors in June.
- Start marketing campaign before peak admission season.
- Prepare more seats or sections.

5. Data Warehouse vs Data Mining

Basis	Data Warehouse	Data Mining
Meaning	Central storage of historical data	Process of finding patterns from data
Purpose	Store and organize data	Discover knowledge
Function	Integrates data	Analyzes data
Output	Reports, dashboards	Patterns, predictions, rules
Example	Student data warehouse	Predict students likely to fail
Role in BI	Foundation	Advanced analysis

Simple relation:

Data Warehouse stores data.

Data Mining discovers knowledge from that data.

6. Data Warehousing and Data Mining Together

They work together in MIS.

Operational Data



Data Warehouse



Data Mining





Example:

Stage	College Example
Operational data	Attendance, marks, LMS login
Data warehouse	Historical student performance database
Data mining	Identify students likely to fail
Knowledge	Low attendance + low LMS login = high risk
Decision	Provide tutorial and counseling

Source	Data
Exam system	Internal marks
Finance system	Fee due
LMS system	Login activity
Counseling system	Student feedback

Step 2: Data Warehouse

Student	Attendance %	Internal Marks	Fee Due	LMS Login/Week	Dropout Status
S1	90	38	No	5	No
S2	62	21	Yes	1	Yes
S3	80	32	No	4	No
S4	55	18	Yes	0	Yes
S5	70	25	No	2	No
S6	58	20	Yes	1	Yes

7. Full MIS Scenario: Student Dropout Prediction

Problem

A college notices some students are dropping out. Management wants to use MIS data to reduce dropout.

Step 1: Data Sources

Source	Data
Attendance system	Attendance percentage

Step 3: Data Mining Pattern

Pattern Found	Meaning
Attendance below 65%	High dropout risk
Internal marks below 22	Academic weakness
Fee due = Yes	Financial pressure
LMS login less than 2	Low engagement

Step 4: Knowledge

Students with:

Attendance below 65%
+ Internal marks below 22
+ Fee due
+ Low LMS login
= Very high dropout risk

Step 5: Decision

Problem	Decision
Low attendance	Call parents and student
Low marks	Extra tutorial class
Fee due	Installment payment plan
Low LMS activity	Academic mentoring

Problem	Decision
High dropout risk	Counseling intervention

Step 6: Impact Measurement

Indicator	Before MIS Action	After MIS Action
Dropout rate	15%	7%
Average attendance	72%	84%
Internal pass rate	78%	90%
LMS activity	45% active	75% active

This shows how data warehousing and data mining support decision-making.

Benefits of Data Warehousing

Benefit	Explanation
Centralized data	All data in one place
Historical analysis	Compare past and present
Better reporting	Faster dashboards
Consistency	Same data for all departments
Strategic planning	Supports long-term decisions

Benefits of Data Mining

Benefit	Explanation
Pattern discovery	Finds hidden relationships
Prediction	Forecasts future outcomes
Customer understanding	Identifies customer segments
Fraud detection	Finds abnormal activities
Competitive advantage	Enables smarter decisions

5.3 Business intelligence tools and techniques

In MIS, BI helps managers answer questions like:

- *What happened?*
- *Why did it happen?*
- *What is happening now?*
- *What may happen in the future?*
- *What decision should we take?*

1. Meaning of Business Intelligence

Business Intelligence is the process of converting raw business data into meaningful information, knowledge, dashboards, reports, and insights for managerial decision-making.

Simple flow:

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Example:

Raw Data	BI Output	Managerial Decision
Daily sales data	Monthly sales dashboard	Increase stock of best-selling product
Customer complaints	Complaint trend report	Improve service department
Student attendance	Risk student report	Provide counseling
Hospital patient flow	Waiting-time dashboard	Add staff in OPD
Website clicks	Conversion analysis	Improve online marketing

2. Why BI Tools Are Needed

Without BI tools, data remains scattered in Excel files, databases, registers, and different systems. BI tools help combine and analyze that data.

Problem Without BI	Solution With BI
Data scattered in many systems	Integrates data in one platform
Manual report preparation	Automated dashboards
Slow decision-making	Faster real-time insights
Hard to identify trends	Visual charts and KPIs

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Problem Without BI	Solution With BI
Data duplication and errors	Data cleaning and validation
Guess-based decision	Evidence-based decision

3. Common Business Intelligence Tools

3.1 Microsoft Excel

Excel is one of the most common BI tools for basic analysis.

Uses

- Data entry
- Sorting and filtering
- Pivot table
- Charts
- Basic statistics
- Dashboard preparation

Example

A sales manager has this data:

Branch	Product	Sales
Butwal	Laptop	300,000
Butwal	Mouse	40,000
Bhairahawa	Laptop	250,000

Branch	Product	Sales
Bhairahawa	Keyboard	60,000

In Excel, the manager can create:

- Pivot table by branch
- Bar chart by product
- Total sales calculation
- Branch-wise comparison

3.2 Microsoft Power BI

Power BI is a professional BI tool used for dashboards and interactive reports.

Uses

- Connect to Excel, SQL, web, cloud, API
- Create interactive dashboards
- Use slicers and filters
- Build KPI cards
- Share reports online

Example Dashboard KPIs

KPI	Value
Total Sales	Rs. 1,500,000
Total Profit	Rs. 420,000
Best Branch	Butwal

KPI	Value
Best Product	Laptop
Customer Growth	18%

Decision:

Management can quickly see which branch and product are performing better.

3.3 Tableau

Tableau is used for advanced and beautiful data visualization.

Uses

- Interactive visualization
- Drag-and-drop dashboard
- Storytelling with data
- Geographic map analysis
- Business trend analysis

Example:

A telecom company can use Tableau to show:

Analysis	Visualization
Customer location	Map
Monthly revenue	Line chart
Package subscription	Bar chart

Analysis	Visualization
Customer churn	Heatmap

3.4 Google Looker Studio

Google Looker Studio is a web-based dashboard tool.

Uses

- Website analytics dashboard
- Google Sheets dashboard
- Marketing report
- Campaign performance report
- Free dashboard sharing

Example:

A digital marketing team can connect Google Analytics data and create a report showing:

- Website visitors
- Page views
- Conversion rate
- Traffic source
- Campaign performance

3.5 SQL

SQL, Structured Query Language, is used to retrieve and analyze data from databases.

Example SQL Query

```
SELECT branch, SUM(sales) AS total_sales
FROM sales_data
GROUP BY branch;
```

Output:

Branch	Total Sales
Butwal	500,000
Bhairahawa	350,000
Palpa	220,000

Decision:

Butwal branch is performing best.

3.6 Python

Python is used for advanced analytics, machine learning, prediction, and automation.

Common Python Libraries

Library	Use
Pandas	Data cleaning and analysis
NumPy	Numerical calculation
Matplotlib	Visualization
Scikit-learn	Machine learning
Seaborn	Statistical visualization
Plotly	Interactive charts

Example use:

- *Predict sales*
 - *Classify customers*
 - *Detect fraud*
 - *Analyze student performance*
 - *Build recommendation system*
-

3.7 R Programming

R is used for statistical analysis and research-based analytics.

Uses

- Statistical modeling
- Forecasting
- Research data analysis
- Visualization
- Survey analysis

Example:

MBA researchers may use R to analyze survey data on customer satisfaction.

3.8 Enterprise BI Tools

Tool	Common Use
SAP BusinessObjects	Enterprise reporting
IBM Cognos	Corporate BI and analytics

Tool	Common Use
Qlik Sense	Interactive BI dashboards
Oracle BI	Enterprise data analysis
SAS BI	Statistical and predictive analytics

4. Business Intelligence Techniques

BI tools are software, while BI techniques are the methods used to analyze and present data.

Technique 1: Reporting

Reporting means preparing structured information from business data.

Types of Reports

Report Type	Example
Daily report	Daily sales report
Monthly report	Monthly profit report
Exception report	Stock below reorder level
Comparative report	This year vs last year
Summary report	Branch-wise total sales

Month	Sales	Expense	Profit
January	500,000	350,000	150,000
February	600,000	390,000	210,000
March	750,000	480,000	270,000

Decision:

Sales and profit are increasing, so business can plan expansion.

Technique 2: Dashboard

A dashboard is a visual screen showing important business indicators.

Dashboard Example

KPI	Value
Total Revenue	Rs. 2,500,000
Total Expense	Rs. 1,700,000
Net Profit	Rs. 800,000
Profit Margin	32%
Best Product	Laptop
Best Branch	Butwal

A dashboard helps managers understand business condition quickly.

Technique 3: KPI Analysis

KPI means **Key Performance Indicator**.

KPI measures whether the organization is achieving its goals.

Examples of KPIs

Business Area	KPI
Sales	Monthly sales growth
Finance	Profit margin
HR	Employee attendance
Education	Pass percentage
Hospital	Patient waiting time
E-commerce	Conversion rate
Inventory	Stock turnover

Example

KPI	Target	Actual	Status
Monthly Sales	1,000,000	1,200,000	Achieved

KPI	Target	Actual	Status
Profit Margin	30%	28%	Needs Improvement
Customer Satisfaction	90%	85%	Needs Improvement
Delivery Time	2 days	3 days	Delayed

Decision:

Sales target is achieved, but customer satisfaction and delivery time need improvement.

Technique 4: OLAP Analysis

OLAP means **Online Analytical Processing**.

It helps analyze data from multiple dimensions.

Example Dimensions

Sales can be analyzed by:

- Product
- Branch
- Month
- Customer type
- Salesperson

OLAP Table Example

Branch	Laptop Sales	Mobile Sales	Accessories	Total
Butwal	500,000	300,000	100,000	900,000

Branch	Laptop Sales	Mobile Sales	Accessories	Total
Bhairahawa	400,000	250,000	80,000	730,000
Palpa	250,000	150,000	50,000	450,000

Manager can drill down:

- Sales by branch
- Sales by product
- Sales by month
- Sales by customer segment

The article also discusses OLAP as an important component of BI for analytical processing and reporting.

Technique 5: Data Visualization

Data visualization presents data using charts, graphs, maps, and dashboards.

Chart Selection

Question	Best Chart
Which product sells most?	Bar chart
How did sales change over time?	Line chart
What is market share?	Pie chart
Which region performs best?	Map chart

Question	Best Chart
Is advertisement related to sales?	Scatter plot
Which branch has high/low performance?	Heatmap

Example

Product	Sales
Laptop	900,000
Mobile	700,000
Mouse	120,000
Keyboard	150,000

A bar chart will clearly show that laptop has the highest sales.

Technique 6: Drill-Down Analysis

Drill-down means moving from summary data to detailed data.

Example

First level:

Year	Sales
2026	12,000,000

Drill down by month:

Month	Sales
January	900,000
February	1,000,000
March	1,300,000

Drill down by branch:

Branch	March Sales
Butwal	600,000
Bhairahawa	450,000
Palpa	250,000

Decision:

March sales increased mainly because Butwal branch performed well.

Technique 7: Slice and Dice

Slice and dice means viewing data from different angles.

Example Dataset

Month	Branch	Product	Sales
Jan	Butwal	Laptop	300,000
Jan	Butwal	Mobile	200,000

Month	Branch	Product	Sales
Jan	Bhairahawa	Laptop	250,000
Feb	Butwal	Laptop	350,000
Feb	Bhairahawa	Mobile	220,000

Slice

Only view Butwal data.

Month	Product	Sales
Jan	Laptop	300,000
Jan	Mobile	200,000
Feb	Laptop	350,000

Dice

View Butwal + Laptop + February.

Month	Branch	Product	Sales
Feb	Butwal	Laptop	350,000

Technique 8: Data Mining

Data mining discovers hidden patterns from large datasets.

Data Mining Techniques

Technique	Use
Classification	Predict category
Clustering	Group similar customers
Association rule	Find items bought together
Regression	Predict numerical value
Anomaly detection	Detect unusual behavior

Example

Retail basket data:

Customer	Items Bought
C1	Rice, Oil
C2	Rice, Oil, Sugar
C3	Bread, Butter
C4	Rice, Oil
C5	Bread, Butter, Jam

Pattern:

Customers who buy rice often buy oil.

Decision:

Place rice and oil together or offer combo discount.

Predictive analytics uses past data to forecast future outcomes.

Example

Month	Sales
January	500,000
February	600,000
March	700,000
April	850,000

Prediction:

May sales may reach around Rs. 950,000.

Decision:

- Increase stock before May.
- Hire temporary staff.
- Prepare marketing campaign.

Technique 10: Prescriptive Analytics

Prescriptive analytics suggests what action should be taken.

Example

Situation	Prescriptive Recommendation
Stock is low and demand is high	Reorder immediately
Customer may leave	Send loyalty offer
Student is high-risk	Provide counseling
OPD waiting time high	Add doctor in peak hours
Delivery delay increasing	Add delivery staff

Predictive analytics says:

What may happen?

Prescriptive analytics says:

What should we do?

5. Types of Analytics in BI

Type	Question Answered	Example
Descriptive analytics	What happened?	Last month sales were Rs. 10 lakh
Diagnostic analytics	Why did it happen?	Sales increased due to festival offer

Type	Question Answered	Example
Predictive analytics	What will happen?	Next month sales may be Rs. 12 lakh
Prescriptive analytics	What should we do?	Increase inventory and marketing

5.4 Big data analytics and its business applications

1. Meaning of Big Data

Big Data refers to extremely large, fast, and complex data that cannot be easily managed by traditional database systems.



Examples:

Source	Big Data Example
Social media	Likes, comments, shares

Source	Big Data Example
E-commerce	Clicks, orders, searches
Banking	ATM, mobile banking, transactions
Hospital	Patient records, lab reports
IoT devices	Sensors, GPS, smart devices
Telecom	Call records, internet usage

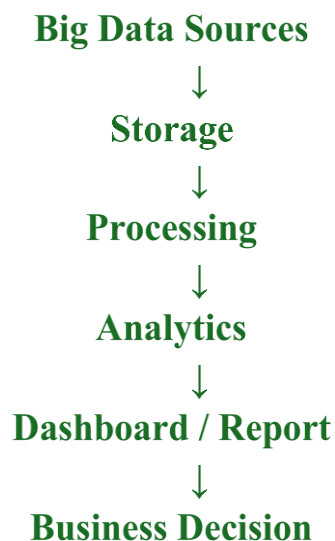
The BI article explains that modern BI systems may include structured, semi-structured, unstructured, streaming, and external data sources for decision-making.

V	Meaning	Example
Volume	Huge amount of data	Millions of transactions
Velocity	Fast data generation	Live online orders
Variety	Different data formats	Text, image, video, database
Veracity	Data quality/trust	Fake reviews, duplicate data
Value	Useful business benefit	Better customer recommendation

3. Big Data Analytics

Big Data Analytics means analyzing large and complex datasets to find patterns, trends, predictions, and useful insights.

Simple flow:



The uploaded article discusses big data architecture with **data sources, storage, batch processing, stream processing, analytical data store, analytics, and reporting.**

4. Batch and Stream Processing

Type	Meaning	Example
Batch Processing	Data processed in groups	Monthly sales report
Stream Processing	Data processed continuously	Live fraud detection

Example:

A bank may use **stream processing** to detect suspicious transactions immediately.

5. Big Data Tools

Tool	Use
Hadoop	Store and process large data
Spark	Fast big data processing
Kafka	Real-time data streaming
MongoDB	Store flexible document data
Cassandra	Distributed big data storage

Tool	Use
Python	Data analysis and machine learning
Power BI/Tableau	Dashboard and visualization



6. Business Applications of Big Data Analytics

A. E-Commerce

Big data helps online businesses understand customer behavior.

Data	Use
Search history	Product recommendation
Cart abandonment	Send discount offer
Reviews	Product improvement
Clicks	Website design improvement

Example decision:

If many customers add products to cart but do not buy, the company can send reminder messages or discount coupons.

B. Banking and Finance

Data	Use
Transaction history	Fraud detection
Loan records	Credit scoring
Customer spending	Personalized banking offer

Example:

If a customer usually transacts in Butwal but suddenly makes a large transaction from another country, the system may mark it suspicious.

C. Healthcare

Data	Use
Patient records	Diagnosis support
Lab reports	Disease prediction
Hospital visit data	Resource planning
Wearable data	Health monitoring

Example:

Hospital can predict peak OPD hours and assign more doctors.

Data	Use
Attendance	Identify irregular students
Exam marks	Find weak students
LMS login	Measure student engagement
Assignment submission	Predict performance

Example:

A college can identify students at dropout risk using attendance, marks, and LMS activity.

Data	Use
GPS data	Route optimization
Delivery time	Performance tracking
Fuel usage	Cost reduction
Traffic data	Faster delivery planning

Example:

A delivery company can choose the shortest route using GPS and traffic data.

E. Retail Business

Data	Use
Sales data	Demand forecasting
Customer data	Loyalty program
Inventory data	Stock planning
Purchase pattern	Combo offers

Example:

If rice and oil are often bought together, the store can create a rice-oil bundle offer.

7. Benefits of Big Data Analytics

Benefit	Explanation
Better decisions	Uses facts and patterns
Faster response	Real-time analysis
Cost reduction	Detects waste and inefficiency
Customer satisfaction	Personalized service
Risk control	Fraud and error detection
Competitive advantage	Smarter and faster business strategy

8. Challenges of Big Data

Challenge	Explanation
Data privacy	Sensitive data must be protected
Data quality	Wrong data gives wrong analysis
High cost	Tools and infrastructure may be expensive

Challenge	Explanation
Skilled manpower	Needs trained analysts
Security risk	Large data can attract cyberattacks
Integration issue	Data comes from many sources